

MILESTONE 2

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PROJECT_ID: CMPG325-2026-136

CLIENT_ID: CLI-136

ORGANISATION: Lenong Panelbeaters (Rustenburg)

INDUSTRY: Automotive

Introduction

This report documents the implementation, configuration and testing of the network designed for Lenong Panelbeaters that is based at Rustenburg, using Cisco Packet Tracer. The focus of Milestone 2 is to implement the approved network design, configure internal DNS services, provide a shared printer solution, enable communication between network segments and verify successful operation through testing.

The network was implemented using VLAN segmentation, inter-VLAN routing and DNS services to support organisational communication and resource sharing. The solution was developed according to the client requirements, design constraint and assigned networking challenge specified in the project brief.

The project brief required the following:

- Design and implement a working network using Cisco Packet Tracer.
- Use the allocated address block 192.168.57.0/24.
- Provide appropriate connectivity and network services.
- Configure and demonstrate DNS (Internal Name Resolution Service).
- Ensure critical services remain available during business hours.
- Implement a shared printer zone serving two departments.

Network Topology

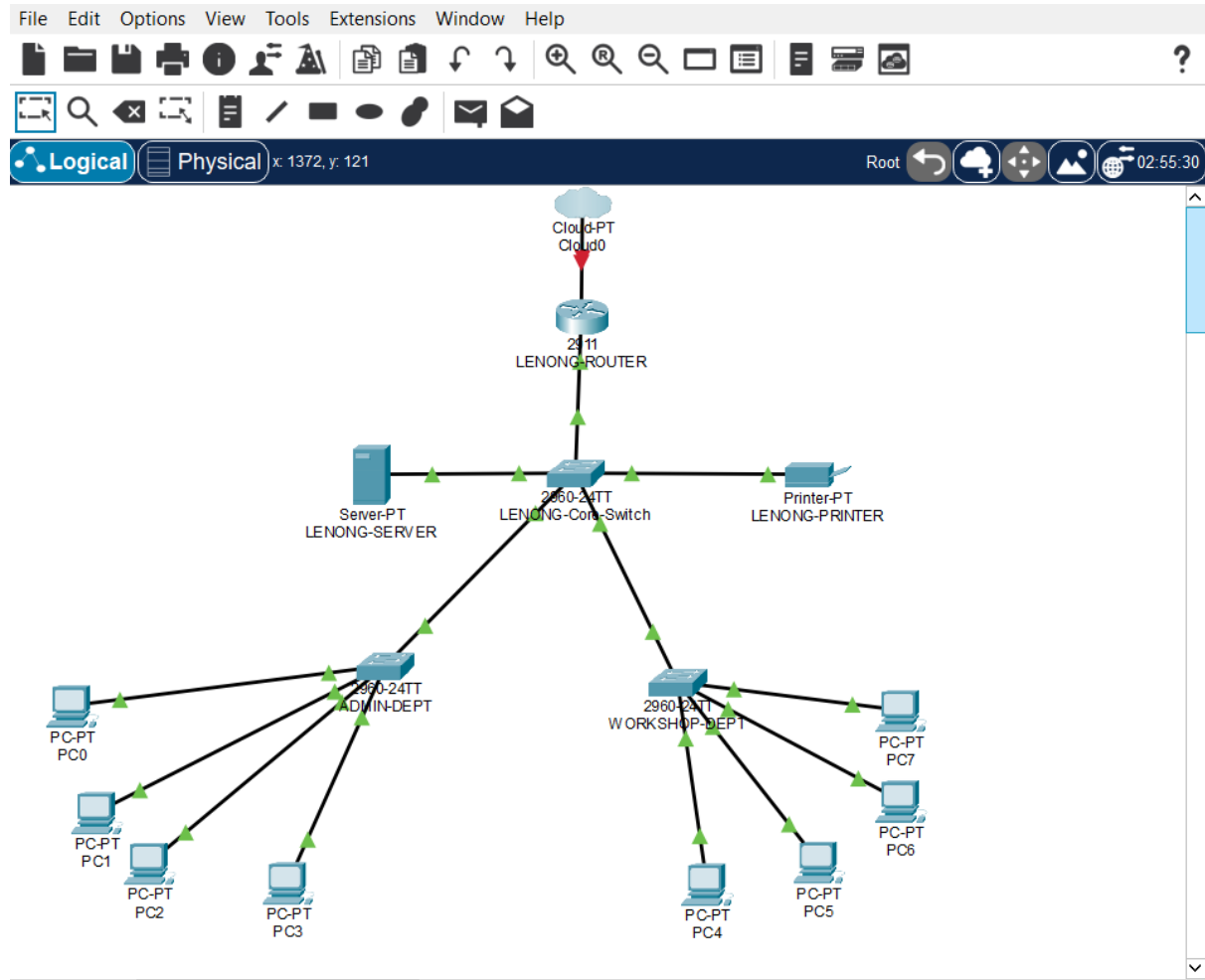
The implemented network follows the approved design developed during Milestone 1. The topology consists of a central core switch connected to departmental switches, a router, a dedicated server and a shared printer.

Devices used:

- Cisco 2911 Router (**LENONG-ROUTER**)
- Cisco 2960-24TT Core Switch (**LENONG-Core-Switch**)
- Cisco 2960-24TT Department A Switch (**ADMIN-DEPT**)
- Cisco 2960-24TT Department B Switch (**WORKSHOP-DEPT**)
- **LENONG-SERVER**
- **LENONG-PRINTER**

- Department A Workstations
- Workshop Department Workstations

FIGURE 1: Network Topology



ORGANISATIONAL DEPARTMENT DESIGN

Administration Department(ADMIN-DEPT) -VLAN 10

The Administration Department represents the front-office operations of Lenong Panelbeaters. This department is responsible for customer interaction, financial administration and daily business operations.

Functions performed within this department include:

- Reception services and customer enquiries.

- Client bookings and job scheduling.
- Financial administration and invoicing.
- Customer records management.
- Communication with insurance companies and suppliers.

The Administration Department was assigned to **VLAN 10** to provide a dedicated and organised network segment for office personnel while maintaining separation from workshop operations.

Administration Workstations

Note that each PC on ADMIN-DEPT is considered a workstation

Workstation	Assigned Staff
ADMIN-PC0-WS01	Reception Personnel
ADMIN-PC1-WS02	Finance & Accounting Personnel
ADMIN-PC2-WS03	Customer Service Personnel
ADMIN-PC3-WS04	Supervisor & Office Manager

Workshop Department(WORKSHOP-DEPT) -VLAN 20

The Workshop Department represents the operational side of Lenong Panelbeaters where vehicle assessments, repairs and maintenance activities are coordinated.

Functions performed within this department include:

- Vehicle damage assessments.
- Mechanical and panel repair activities.
- Parts ordering and inventory tracking.
- Workshop job management.
- Repair status updates and reporting.

The Workshop Department was assigned to VLAN 20 to separate workshop operations from administrative functions while allowing controlled communication with shared resources.

Workshop workstation

Note that each PC on WORKSHOP-DEPT is considered a workstation

Workstation	Assigned Staff
WORKSHOP-PC4-WS01	Vehicle Assessment Personnel
WORKSHOP -PC5-WS02	Repair Technicians
WORKSHOP -PC6-WS03	Parts & Inventory Personnel
WORKSHOP -PC7-WS04	Workshop Supervisor

LENONG-SERVER

The server LENONG-SERVER was deployed within the Server VLAN to provide critical network services. The server hosts the internal DNS service and represents the central platform for organisational resources.

The server supports:

- DNS name resolution.
- File service availability.
- Application service availability.
- Shared resource management.

LENONG-PRINTER

The shared printer LENONG-PRINTER was deployed within the Printer VLAN to satisfy Change Request CR8.

The printer is accessible by both the Administration and Workshop departments through inter-VLAN routing.

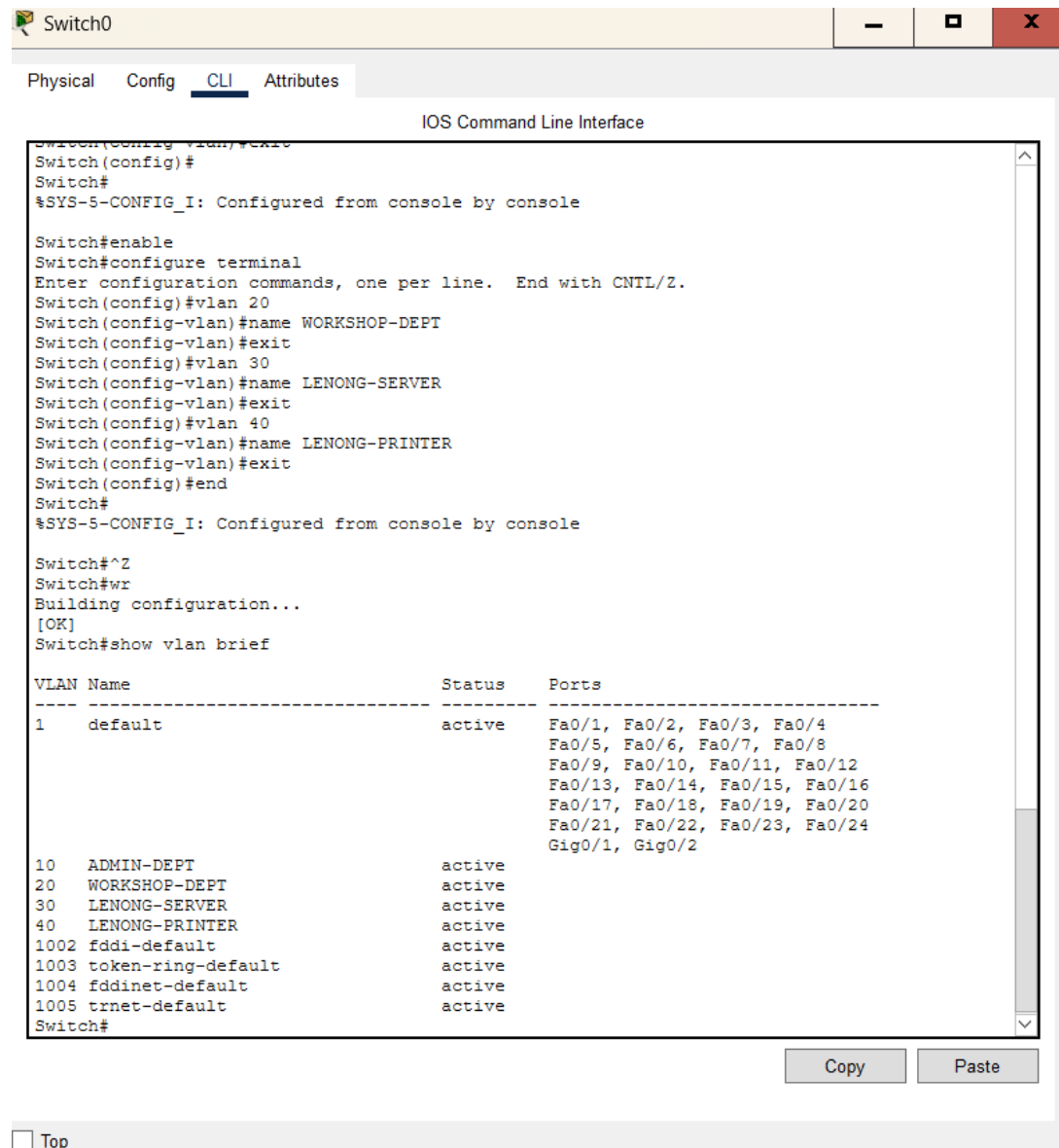
This implementation allows staff from different business areas to utilise a single shared printing resource while maintaining logical network separation.

Cloud0

The topology includes a Cloud device representing a potential external network or Internet connection. However, Internet connectivity was not required as part of the project scope. The primary focus of the implementation was internal network communication, DNS name resolution, inter-VLAN routing and shared resource access.

Cloud0 was retained within the topology to represent a potential future Internet or ISP connection. However, external connectivity was not required within the scope of this milestone and was therefore not configured.

Figure 2a: VLAN creating on LENONG-Core-Switch(Switch0)



The screenshot shows a network switch interface with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the IOS Command Line Interface. The configuration process includes enabling the switch, entering terminal mode, and creating four VLANs: 20 (WORKSHOP-DEPT), 30 (LENONG-SERVER), 40 (LENONG-PRINTER), and 10 (ADMIN-DEPT). The configuration is saved with the 'wr' command. A 'show vlan brief' command is executed, resulting in a table of VLANs.

```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Switch(config)#enable
Switch(config)#
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name WORKSHOP-DEPT
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name LENONG-SERVER
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name LENONG-PRINTER
Switch(config-vlan)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#^Z
Switch#wr
Building configuration...
[OK]
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2

10   ADMIN-DEPT              active
20   WORKSHOP-DEPT           active
30   LENONG-SERVER            active
40   LENONG-PRINTER           active
1002 fddi-default           active
1003 token-ring-default     active
1004 fddinet-default        active
1005 trnet-default          active
Switch#
```

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Top

The VLAN design provides logical separation between departments while still allowing controlled communication through inter-VLAN routing.

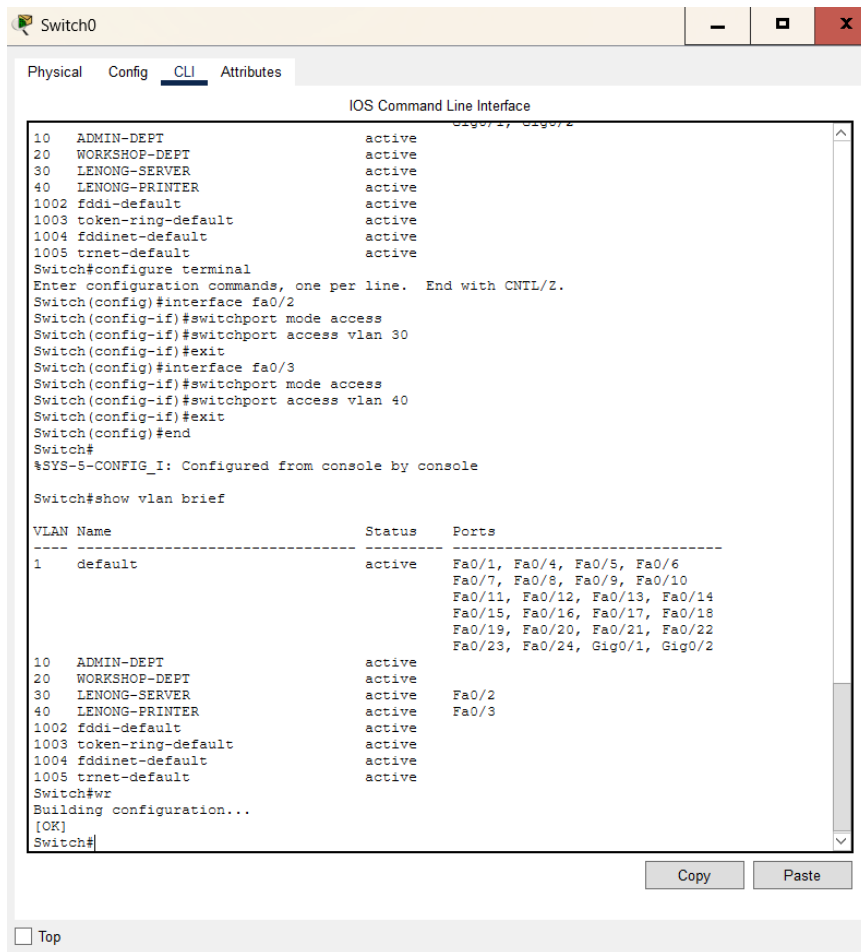


Figure 2b: Configure ADMIN-DEPT On LENONG-Core-Switch(Switch0)

The core switch was configured to support VLAN segmentation and provide connectivity to all network devices.

Access ports were assigned to the appropriate VLANs while trunk ports were configured to transport VLAN traffic between switches and the router.

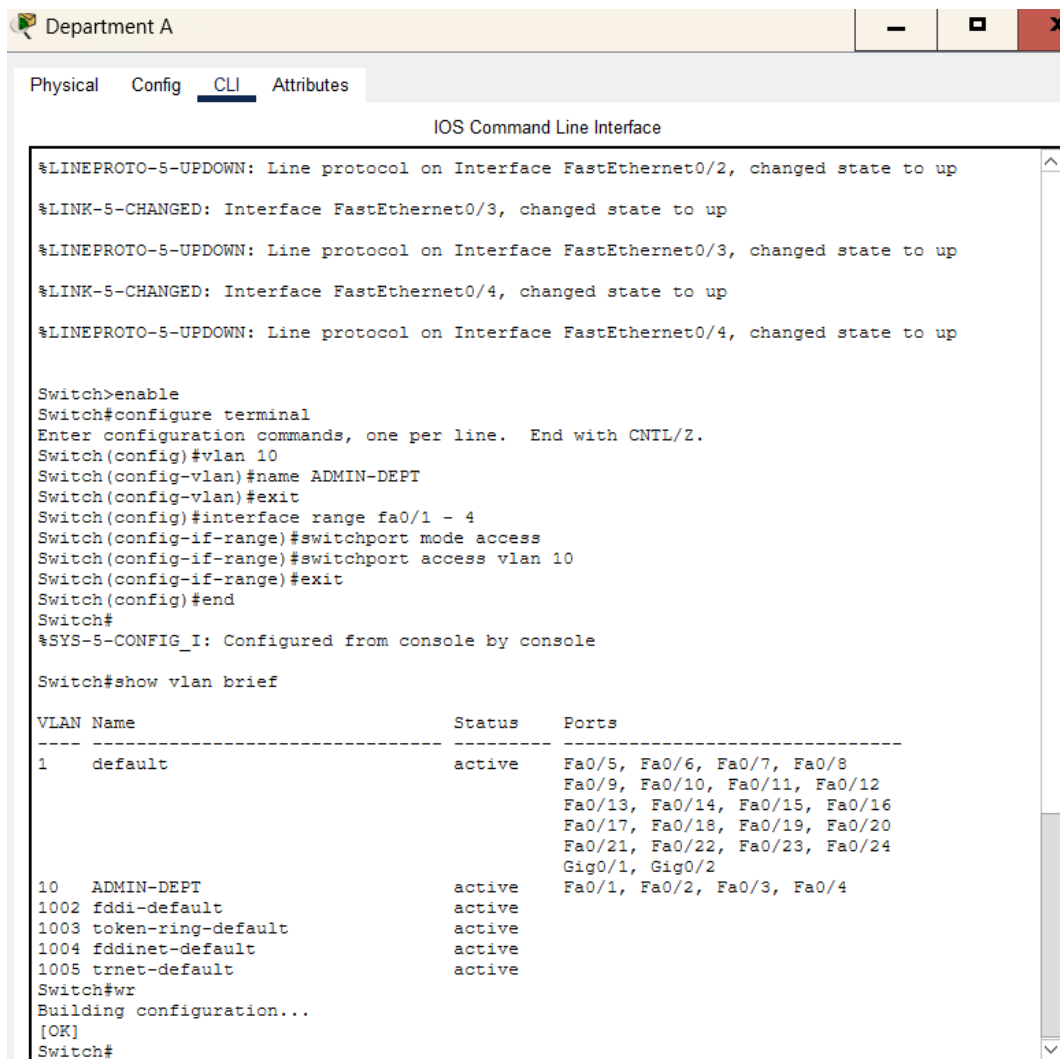


Figure 3: Configure ADMIN-DEPT (DEPT-A)

Department B

Physical Config CLI Attributes

IOS Command Line Interface

```

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name WORKSHOP-DEPT
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1 - 4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
20	WORKSHOP-DEPT	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```

Switch#

```

Figure 4: Configure WORKSHOP-DEPT (DEPT-B)

Figure 5: TRUNK LINKS

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                Status    Ports
----
1    default                active    Fa0/1, Fa0/4, Fa0/5, Fa0/6
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24
10   ADMIN-DEPT             active
20   WORKSHOP-DEPT          active
30   LENONG-SERVER          active    Fa0/2
40   LENONG-PRINTER         active    Fa0/3
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active

Switch#wr
Building configuration...
[OK]
Switch#show interface trunk

Port      Mode      Encapsulation  Status        Native vlan
Gig0/1    on        802.1q         trunking      1
Gig0/2    on        802.1q         trunking      1

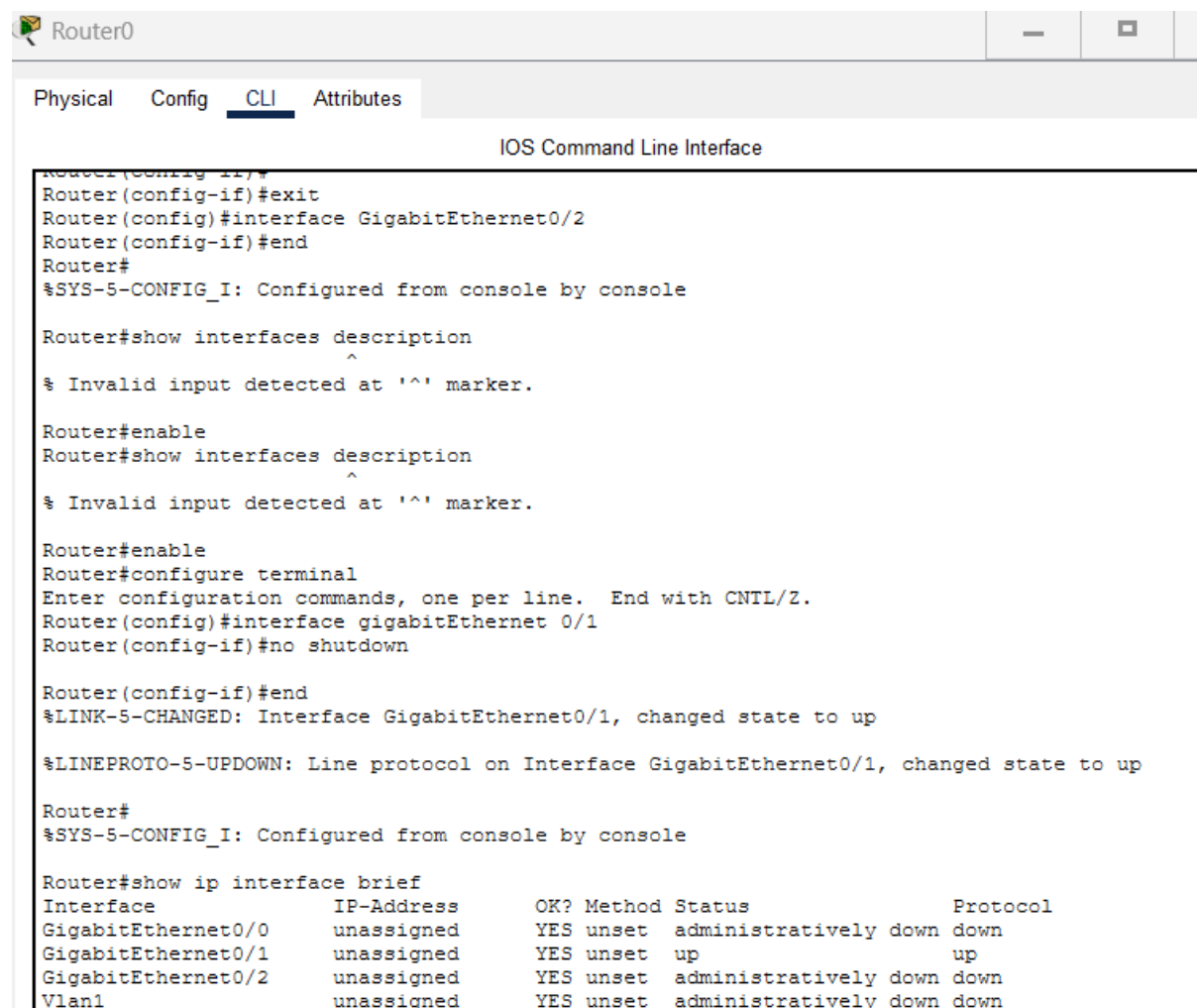
Port      Vlans allowed on trunk
Gig0/1    1-1005
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20,30,40
Gig0/2    1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,20,30,40
Gig0/2    1,10,20,30,40

Switch#
```

Figure 6: Router Configuration



Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#  
Router(config-if)#exit  
Router(config)#interface GigabitEthernet0/2  
Router(config-if)#end  
Router#  
%SYS-5-CONFIG_I: Configured from console by console  
  
Router#show interfaces description  
^  
% Invalid input detected at '^' marker.  
  
Router#enable  
Router#show interfaces description  
^  
% Invalid input detected at '^' marker.  
  
Router#enable  
Router#configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#interface gigabitEthernet 0/1  
Router(config-if)#no shutdown  
  
Router(config-if)#end  
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up  
  
Router#  
%SYS-5-CONFIG_I: Configured from console by console  
  
Router#show ip interface brief  
Interface IP-Address OK? Method Status Protocol  
GigabitEthernet0/0 unassigned YES unset administratively down down  
GigabitEthernet0/1 unassigned YES unset up up  
GigabitEthernet0/2 unassigned YES unset administratively down down  
Vlan1 unassigned YES unset administratively down down
```

Figure 7:Trunk Configuration

The screenshot shows a network switch window titled "Switch0" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface".

The CLI output shows the following configuration and status:

```
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false

Switch#show interfaces fastEthernet 0/1 status
^
% Invalid input detected at '^' marker.

Switch#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1
Gig0/1    on        802.1q         trunking    1
Gig0/2    on        802.1q         trunking    1

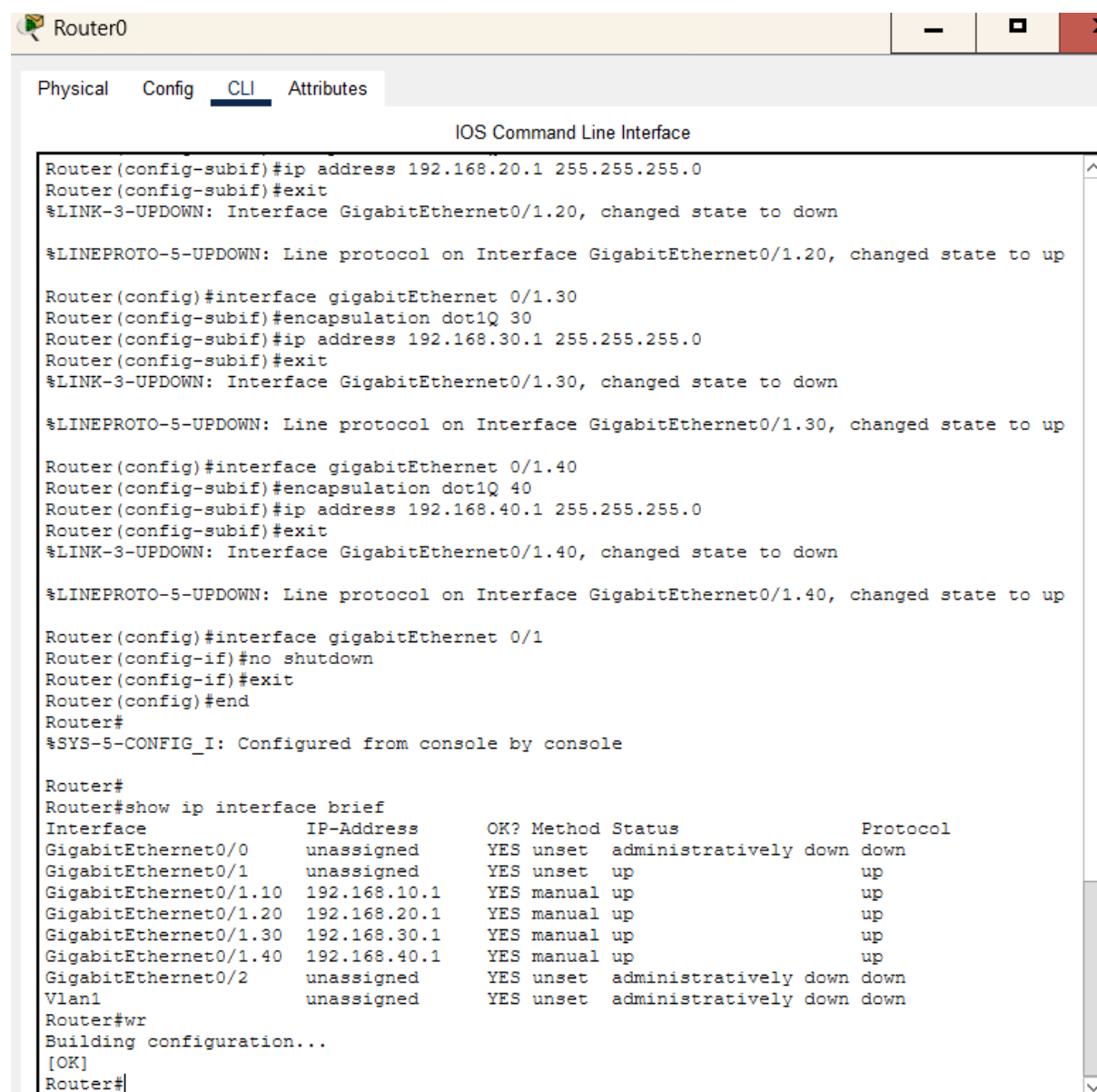
Port      Vlans allowed on trunk
Fa0/1     1-1005
Gig0/1    1-1005
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30,40
Gig0/1    1,10,20,30,40
Gig0/2    1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30,40
Gig0/1    1,10,20,30,40
Gig0/2    1,10,20,30,40

Switch#
```

Figure 8:Router on stick configuration



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.20, changed state to up

Router(config)#interface gigabitEthernet 0/1.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#exit
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.30, changed state to up

Router(config)#interface gigabitEthernet 0/1.40
Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#exit
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.40, changed state to down

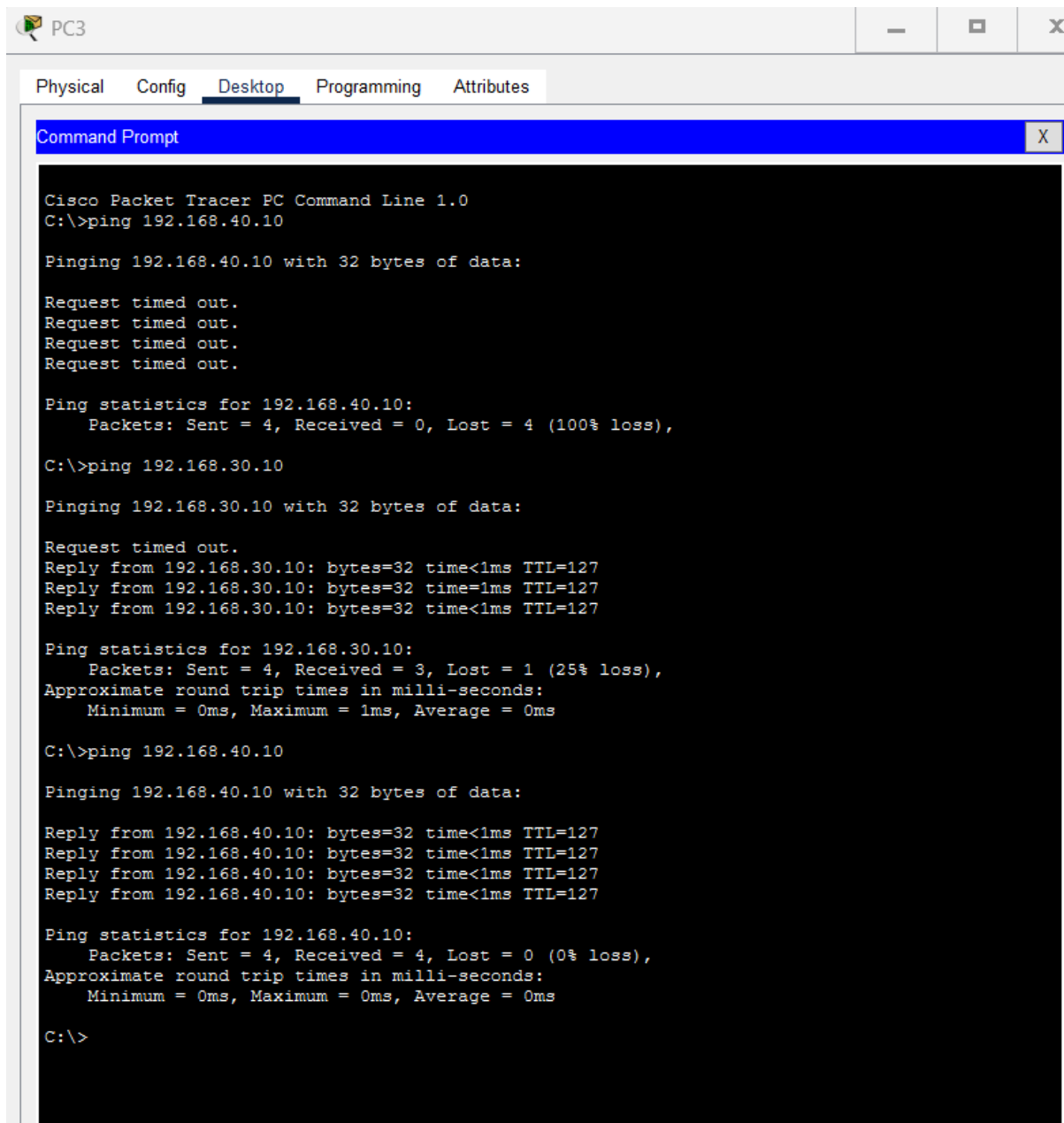
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.40, changed state to up

Router(config)#interface gigabitEthernet 0/1
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0 unassigned      YES unset  administratively down  down
GigabitEthernet0/1 unassigned      YES unset  up              up
GigabitEthernet0/1.10 192.168.10.1    YES manual  up              up
GigabitEthernet0/1.20 192.168.20.1    YES manual  up              up
GigabitEthernet0/1.30 192.168.30.1    YES manual  up              up
GigabitEthernet0/1.40 192.168.40.1    YES manual  up              up
GigabitEthernet0/2 unassigned      YES unset  administratively down  down
Vlan1              unassigned      YES unset  administratively down  down
Router#wr
Building configuration...
[OK]
Router#
```

Router sub interfaces were configured to provide inter-VLAN communication between departmental VLANs and service VLANs.

Figure 9: Successful Printer ping



```
PC3
Physical Config Desktop Programming Attributes
Command Prompt X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

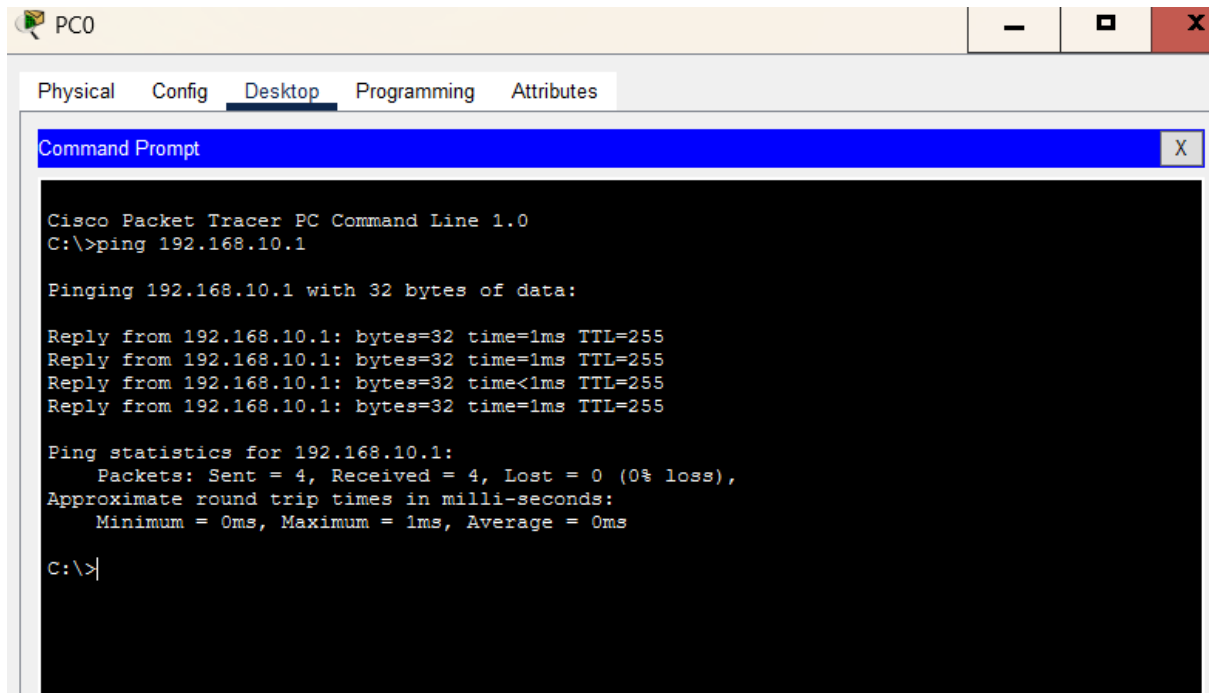
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

VLAN 40 was implemented as a shared printer zone, allowing both ADMIN-DEPT (Department A) and WORKSHOP-DEPT (Department B) users to access the shared printer through inter-VLAN routing. This satisfies Change Request CR8.

Figure 10: Admin gateway ping



The screenshot shows a Cisco Packet Tracer window for PC0. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of a ping command to 192.168.10.1, which is successful. The output includes details about the data size (32 bytes), response times (1ms), and TTL (255). Ping statistics show 4 packets sent and received with 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

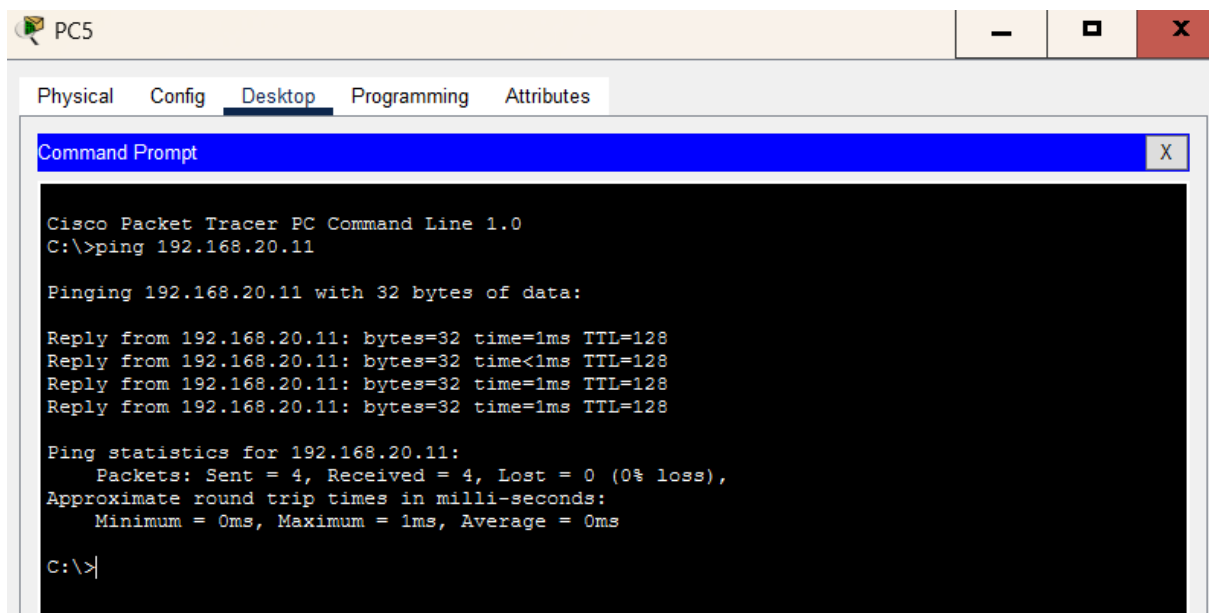
Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figure 11: WORKSHOP ping



The screenshot shows a Cisco Packet Tracer window for PC5. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of a ping command to 192.168.20.11, which is successful. The output includes details about the data size (32 bytes), response times (1ms), and TTL (128). Ping statistics show 4 packets sent and received with 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time=1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time=1ms TTL=128
Reply from 192.168.20.11: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figure 12a: DNS Configuration

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type **A Record** ▼

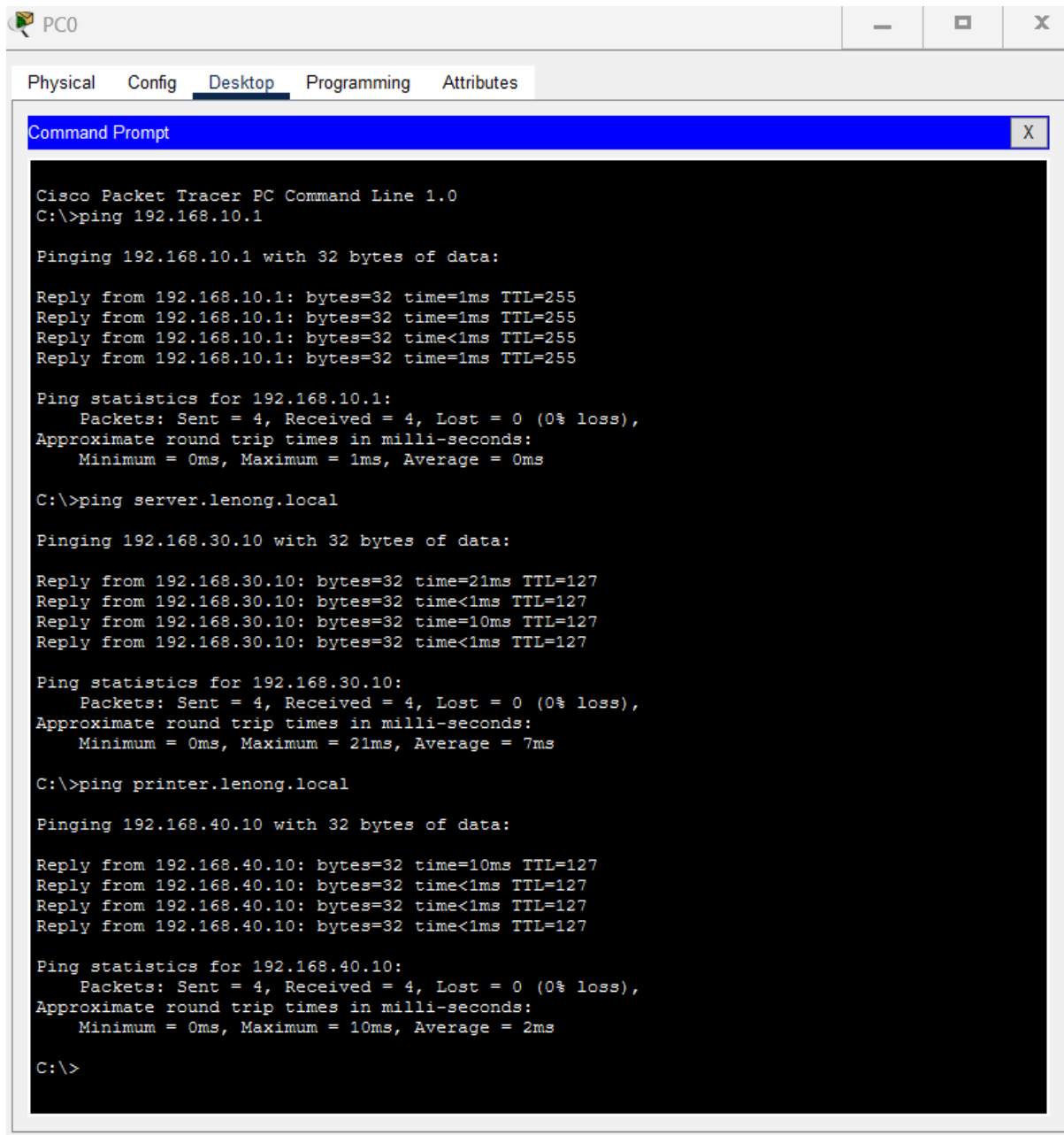
Address

No.	Name	Type	Detail
0	printer.lenong.local	A Record	192.168.40.10
1	server.lenong.local	A Record	192.168.30.10

The DNS service was configured on **LENONG-SERVER** to provide internal hostname resolution within the organisation.

The DNS server allows users to access network resources using meaningful hostnames instead of remembering numerical IP addresses.

DNS records were created for key resources within the organisation.



The screenshot shows a Cisco Packet Tracer PC Command Prompt window. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, and the Command Prompt window is open. The text in the Command Prompt is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping server.lenong.local

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time=21ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=10ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 21ms, Average = 7ms

C:\>ping printer.lenong.local

Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.40.10: bytes=32 time=10ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
```

Figure 13: DNS Service

The DNS challenge assigned in the project brief was successfully implemented through the creation of internal DNS resource records. Users can now access organisational resources using hostnames such as *server.lenong.local* and *printer.lenong.local* rather than remembering numerical IP addresses.

Design Constraint Implementation

The project brief specified that critical services such as file, print and application services must remain available during business hours. This requirement was addressed by deploying critical resources within dedicated VLANs and ensuring they could be accessed through inter-VLAN routing. The implementation of **LENONG-SERVER** and **LENONG-PRINTER** in separate service networks improves network organisation and supports the availability of essential business services.

CONCLUSION

This project successfully designed, implemented and tested a computer network solution for Lenong Panelbeaters (Rustenburg). The network was developed using Cisco Packet Tracer and was configured to meet the client requirements specified in the project brief, including the implementation of DNS as the assigned networking challenge and the provision of a shared printer zone serving multiple departments.

The final network design incorporated VLAN segmentation to separate the Administration and Workshop departments while maintaining secure communication through inter-VLAN routing. Critical resources, including **LENONG-SERVER** and **LENONG-PRINTER**, were deployed within dedicated network segments to support the organisation's operational requirements and ensure access to essential services during business hours.

The internal DNS service was successfully configured on **LENONG-SERVER**, allowing users to access network resources through meaningful hostnames rather than IP addresses. The implementation of a dedicated printer zone also satisfied Change Request CR8 by allowing both the Administration and Workshop departments to access the shared printing service.

Comprehensive testing confirmed successful communication between workstations, servers and shared resources. DNS name resolution, inter-VLAN routing and printer accessibility were all verified and demonstrated to operate as intended. The completed solution therefore meets the technical, operational and organisational requirements of Lenong Panelbeaters and provides a reliable foundation for the company's day-to-day business activities.

This project provided practical experience in VLAN implementation, inter-VLAN routing, DNS configuration and network troubleshooting using Cisco Packet Tracer. It reinforced the importance of network segmentation, resource sharing and service availability within a business environment. The project also improved practical skills in testing and verifying network functionality.

